

Time Series

Decomposition : Trend Estimation

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Time Series Course 1GAMMA

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Session Objectives

By the end of this session, you will be able to :

- Understand time series decomposition
- Understand the trend component
- Estimate and remove trend from time series
- Apply moving average methods
- Apply parametric regression methods
- Understand non-parametric regression methods
- Select appropriate trend estimation methods

Time Series : Basics

Time series data :

$$Y_t, \quad t = 1, 2, \dots, T$$

Model decomposition :

$$Y_t = T_t + S_t + \varepsilon_t$$

where :

- T_t : trend component
- S_t : seasonal component
- ε_t : stationary component

Goal :

- Estimate T_t and S_t
- Remove them to obtain stationary process

Trend Estimation

Goal : eliminate trend component

Approaches :

- Estimate trend and subtract it
- Difference the data

Methods :

- Moving Average
- Parametric Regression
- Non-Parametric Regression

Trend Estimation : Moving Average

Trend estimate :

$$\hat{T}_t = \frac{1}{2d+1} \sum_{i=-k}^k Y_{t+i}$$

where :

- d : window size
- used to smooth time series

Properties :

- Simple
- Effective smoothing
- Removes noise

Example : Moving Average in R

Example : Estimate trend using moving average

```
# Generate example time series
t <- 1:100
y <- 0.5*t + rnorm(100,0,5)

# Moving average with window size 5
trend <- filter(y, rep(1/5,5), sides=2)

# Plot
plot(t,y,type="l",col="gray",
      main="Moving Average Trend Estimation")
lines(t,trend,col="blue",lwd=2)
```

Example : Moving Average in Python

```
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd

# Generate time series
t = np.arange(100)
y = 0.5*t + np.random.normal(0,5,100)

# Moving average
trend = pd.Series(y).rolling(window=5).mean()

# Plot
plt.plot(t,y,color='gray')
plt.plot(t,trend,color='blue',linewidth=2)
plt.title("Moving Average Trend Estimation")
plt.show()
```

Trend Estimation : Parametric Regression

Assume polynomial model :

$$T_t = \beta_0 + \beta_1 t + \beta_2 t^2 + \dots + \beta_p t^p$$

Estimation using regression :

- Linear trend ($p = 1$)
- Quadratic trend ($p = 2$)

Model selection used to choose order.

Example : Linear Trend Estimation in R

```
# Linear regression
model <- lm(y ~ t)

# Estimated trend
trend <- predict(model)

# Plot
plot(t,y,type="l",col="gray")
lines(t,trend,col="red",lwd=2)
```

Example : Linear Trend Estimation in Python

```
from sklearn.linear_model import LinearRegression

# reshape for sklearn
X = t.reshape(-1,1)

model = LinearRegression()
model.fit(X,y)

trend = model.predict(X)

plt.plot(t,y,color='gray')
plt.plot(t,trend,color='red',linewidth=2)
plt.title("Linear Trend Estimation")
plt.show()
```

Trend Estimation : Non-Parametric Regression

Kernel regression :

$$\hat{T}_t = \sum w_i Y_i$$

Local polynomial regression :

- Fit polynomial locally
- Flexible
- Accurate

Other methods :

- Splines
- Wavelets

Preferred methods :

- Local polynomial regression
- Splines regression

Advantages :

- Better accuracy
- Handles boundaries well
- Flexible modeling

Choice depends on smoothness of trend.

Example : Local Regression (LOESS) in R

Local regression estimates the trend using local polynomial fitting.

```
# Generate example time series
t <- 1:100
y <- 0.5*t + 10*sin(t/10) + rnorm(100,0,5)
# Local regression using LOESS
model <- loess(y ~ t, span=0.2)
# Estimated trend
trend <- predict(model)
# Plot results
plot(t, y, type="l", col="gray",
      main="Local Regression (LOESS) Trend Estimation",
      xlab="Time", ylab="Value")
lines(t, trend, col="blue", lwd=2)
legend("topleft",
       legend=c("Original Series", "LOESS Trend"),
       col=c("gray", "blue"), lwd=2)
```

Example : Non-Parametric Trend (Python)

```
from statsmodels.nonparametric.smoothers_lowess import lowess

trend = lowess(y,t,frac=0.1)

plt.plot(t,y,color='gray')
plt.plot(trend[:,0],trend[:,1],
         color='green',linewidth=2)
plt.title("LOWESS Trend Estimation")
plt.show()
```

Summary

In this session, we covered :

- Time series decomposition
- Trend component
- Moving average estimation
- Parametric regression
- Non-parametric regression